AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems
Final Examination

Summer Semester 2024/2025
Date: 3/Sep./2025

Number of Exam Pages: 4 (including this cover sheet)	Time Allowed: 2 Hours

Instructions:

- Total Marks:**50**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **not** allowed.

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Part 1: Multiple Choice Questions: [10 marks]

Question 1: Choose the most correct answer for each of the following:

1. Device controller informs CPU that it has finished its operation by causing a/an:
   a) Request
   b) Semaphore signal()
   c) interrupt
   d) Mutex release()
2. ____ is time it takes for the dispatcher to stop one process and start another run.
   a) Dispatch loading
   b) Dispatch handling
   c) Dispatch response time
   d) Dispatch latency.
3. The correct sequence for protecting a critical section using binary **Semaphore** is:
   a) signal() followed by wait()
   b) release() followed by acquire()
   c) wait() followed by signal()
   d) acquire() followed by release()
4. What is the primary purpose of demand paging?
   a) To reduce the time a program takes to execute.
   b) To load all pages of a process into memory at once.
   c) To load pages only when they are needed.
   d) To manage CPU scheduling.
5. A significant problem with priority scheduling algorithms is:
   a) Knowing the next burst
   b) Starvation
   c) Knowing the time quantum
   d) Complexity
6. The number of processes that are completed per time unit is the:
   a) Turnaround time
   b) CPU utilization
   c) Throughput
   d) Response time.
7. Which scheduling algorithm is best suited for time-sharing systems?
   a) Round-Robin
   b) First-Come, First-Served
   c) Shortest Job Next
   d) Priority Scheduling.
8. If a system is in the unsafe mode, this means that we have:
   a) Deadlock
   b) A possibility of a deadlock
   c) No deadlock
   d) Either a or b
9. Which of the following is not a core function of an operating system?
   a) File management
   b) Process management
   c) Web browsing
   d) Memory management.
10. ____ is used for managing access to shared data between processes.
    a) POSIX
    b) Debugger
    c) Locks
    d) Security

Part 2: Short and Essay questions: [15 marks]

Question 2: [3 marks]

Multiprogramming offers several advantages, including increased CPU utilization and improved memory utilization. Briefly explain how multiprogramming achieves each of these benefits.

Question 3: [4 marks]

Write the pseudo-code for the **acquire()** and **release()** operations of a mutex, using a global Boolean variable **available** to ensure mutual exclusion.

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Question 4: [4 marks]

In OS context, PCB is responsible for controlling the states of the processes.

a- What does the acronym PCB stand for? (1 mark)

b- List three information, other than process state, associated with each process and found in the PCB. (3 marks)

Question 5: [4 marks]

a) Discuss how deadlocks can be **avoided**. (3 marks)

b) Draw the diagram that shows the safe, unsafe, and the deadlock state. (1 mark)

Part 3: Problem Solving questions: [25 marks]

Question 6: [6 marks]

Assume having three processes P1, and P2 and the following resources:

- One instance of R1
- Two instances of R2
- One instance of R3

Moreover, below is the relation between the resources and the processes:

- P1 holds one instance of R1 and R2 and waiting for an instance of R3,
- P2 holds one instance of R2 and R3 and waiting for R1.

a) Draw the corresponding Resource-Allocation Graph (RAG). (3 marks)

b) Determine if the RAG contains a cycle and indicate it. (1 mark)

c) Deduce if there is a deadlock. Justify your answer based on the rule. (2 marks)

Question 7: [6 marks]

Consider the following segment table:

Segment	Base	Limit
0	250	240
1	500	77
2	100	120
3	800	96

Find the physical addresses for the following logical addresses and deduce whether this address will be of legal reference or not. (Show the calculations)

1. Segment: 0, Logical address: 230 (2 marks)
2. Segment: 1, Logical address: 80 (2 marks)
3. Segment: 3, Logical address: 95 (2 marks)

Question 8: [6 marks]

Consider the following page reference string: 7,5,6,3,7,3,6,5,2,8

Ref String	7	5	6	3	7	3	6	5	2	8
f1										
f2										
f3										
H/PF										

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Assume that there are **three frames**, and the **FIFO** was used.
Answer the following questions:

a) Fill in the table above after copying it to your answer booklet. (5 marks)
b) Find the number of Page Faults (PF). (1 mark)

N.B: In the last row, you should indicate whether we have a hit (**H**) or a page fault (**PF**)

Question 9: [7 marks]

Consider the memory partitions of 400 KB, 100 KB, 600 KB, 300 and 200 KB, and processes of 80 KB, 250 KB, 500 KB, and 250 KB (arrives in order).

a) How would each of the first-fit, best-fit and worst-fit algorithms place the above processes in the given memory partitions. (6 marks)
b) Which algorithm makes the most efficient use of memory? (1 mark)

N.B: Use the table below and fill it for each algorithm.

Memory Partition	Process
400	
100	
600	
300	
200	

End of questions

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